Flow through Pipes

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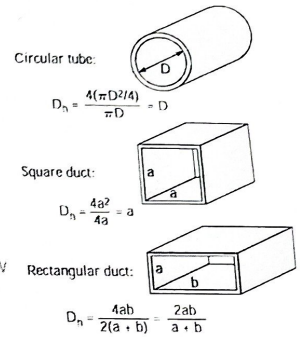
Reynolds Number

- For circular pipes

$$D_h = \frac{4A_c}{p} = \frac{4(\pi D^2/4)}{\pi D} = D$$

- For flow in circular pipes

$Re \leq 2300$	laminar flow
$2300 \leq Re \leq 4000$	transitional flow
$Re \geq 4000$	turbulent flow

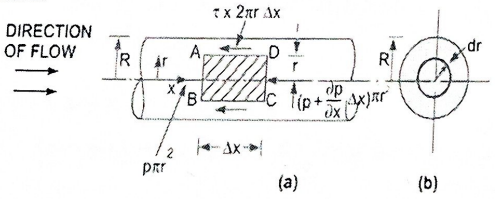


Flow through pipe

- Consider a flow through pipe having radius R and and small element ABCD at distance r, length of element is Δx
- P is intensity of pressure on AB and

$$R_e = \frac{\rho VD}{\mu}$$

where ρ = Density of fluid flowing through pipe
V = Average velocity of fluid
D = Diameter of pipe and
μ = Viscosity of fluid.



1. The pressure force, $p \times \pi r^2$ on face AB.
2. The pressure force, $\left(p + \frac{\partial p}{\partial x}\Delta x\right)\pi r^2$ on face CD.
3. The shear force, $\tau \times 2\pi r \Delta x$ on the surface of fluid element. As there is no acceleration, hence the summation of all forces in the direction of flow must be zero *i.e.*,

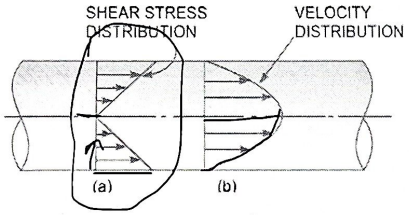
$$p\pi r^2 - \left(p + \frac{\partial p}{\partial x}\Delta x\right)\pi r^2 - \tau \times 2\pi r \times \Delta x = 0$$

or
$$-\frac{\partial p}{\partial x}\Delta x \pi r^2 - \tau \times 2\pi r \times \Delta x = 0$$

or
$$-\frac{\partial p}{\partial x} \cdot r - 2\tau = 0$$

$\therefore$
$$\tau = -\frac{\partial p}{\partial x}\frac{r}{2}$$

The shear stress τ across a section varies with 'r' as $\frac{\partial p}{\partial x}$ across a section is constant. Hence shear stress distribution across a section is linear



$$\therefore \quad 0 = \frac{1}{4\mu}\frac{\partial p}{\partial x} R^2 + C$$

$$\therefore \quad C = -\frac{1}{4\mu}\frac{\partial p}{\partial x} R^2$$

Substituting this value of C in equation (9.2), we get

$$u = \frac{1}{4\mu}\frac{\partial p}{\partial x} r^2 - \frac{1}{4\mu}\frac{\partial p}{\partial x} R^2$$

$$= -\frac{1}{4\mu}\frac{\partial p}{\partial x}[R^2 - r^2]$$

In equation values of μ, $\frac{\partial p}{\partial x}$ and R are constant, which means the velocity, u varies with the square of r. Thus equation is a equation of parabola. This shows that the velocity distribution across the section of a pipe is parabolic.

Velocity Distribution. To obtain the velocity distribution across a section, the value of shear stress $\tau = \mu \frac{du}{dy}$ is substituted in equation

But in the relation $\tau = \mu \frac{du}{dy}$, y is measured from the pipe wall. Hence

$$y = R - r \quad \text{and} \quad dy = -dr$$

$$\therefore \quad \tau = \mu \frac{du}{-dr} = -\mu \frac{du}{dr}$$

Substituting this value

$$-\mu \frac{du}{dr} = -\frac{\partial p}{\partial x}\frac{r}{2} \quad \text{or} \quad \frac{du}{dr} = \frac{1}{2\mu}\frac{\partial p}{\partial x} r$$

Integrating this above equation w.r.t. 'r', we get

$$u = \frac{1}{4\mu}\frac{\partial p}{\partial x} r^2 + C$$

where C is the constant of integration and its value is obtained from the boundary condition that at $r = R$, $u = 0$.

Ratio of Maximum Velocity to Average Velocity. The velocity is maximum, when $r = 0$ in equation Thus maximum velocity, U_{max} is obtained as

$$U_{max} = -\frac{1}{4\mu}\frac{\partial p}{\partial x} R^2$$

$$\frac{U_{max}}{\bar{u}} = \frac{-\frac{1}{4\mu}\frac{\partial p}{\partial x} R^2}{\frac{1}{8\mu}\left(-\frac{\partial p}{\partial x}\right) R^2} = 2.0$$

$\therefore$ Ratio of maximum velocity to average velocity = 2.0.

Pressure drop over Length – Hagen Poiseuille Formula

$$\bar{u} = \frac{1}{8\mu}\left(\frac{-\partial p}{\partial x}\right)R^2 \quad \text{or} \quad \left(\frac{-\partial p}{\partial x}\right) = \frac{8\mu\bar{u}}{R^2}$$

Integrating the above equation w.r.t. x, we get

$$-\int_1^2 dp = \int_1^2 \frac{8\mu\bar{u}}{R^2}\,dx$$

$$\therefore \quad -[p_1 - p_2] = \frac{8\mu\bar{u}}{R^2}[x_1 - x_2] \text{ or } (p_1 - p_2) = \frac{8\mu\bar{u}}{R^2}[x_2 - x_1]$$

$$= \frac{8\mu\bar{u}}{R^2}L \qquad \{\because \; x_2 - x_1 = L$$

$$= \frac{8\mu\bar{u}L}{(D/2)^2} \qquad \left\{\because \; R = \frac{D}{2}\right\}$$

or $$(p_1 - p_2) = \frac{32\mu\bar{u}L}{D^2}, \quad \text{where } p_1 - p_2 \text{ is the drop of pressure.}$$

$\therefore$ Loss of pressure head $= \dfrac{p_1 - p_2}{\rho g}$

$\therefore$ $$\frac{p_1 - p_2}{\rho g} = h_f = \frac{32\mu\bar{u}L}{\rho g D^2}$$

Equation is called **Hagen Poiseuille Formula.**

- *Kinetic Energy and Momentum correction factor*

Kinetic energy correction factor is defined as the ratio of the kinetic energy of the flow per second based on actual velocity across a section to the kinetic energy of the flow per second based on average velocity

$$\alpha = \frac{\int_A v^3\,dA}{V^3 A}$$

where

- v = local velocity,
- V = average velocity,
- A = cross-sectional area.
- If flow is uniform → $\alpha = 1$.
- For laminar flow in a pipe → $\alpha = 2.0$.
- For turbulent flow in a pipe → $\alpha \approx 1.03$–1.06.

Momentum Correction Factor. It is defined as the ratio of momentum of the flow per second based on actual velocity to the momentum of the flow per second based on average velocity across a section. It is denoted by β. Hence mathematically,

$$\beta = \frac{\text{Momentum per second based on actual velocity}}{\text{Momentum per second based on average velocity}}.$$

$$\beta = \frac{\int_A v^2\,dA}{V^2 A}$$

- If flow is uniform → $\beta = 1$.
- For laminar flow in a pipe → $\beta = 1.33$.
- For turbulent flow in a pipe → $\beta \approx 1.01$–1.03.

Expression for Loss of Head Due to Friction in Pipes. Consider a uniform horizontal pipe, having steady flow as shown in Fig. . Let 1-1 and 2-2 are two sections of pipe.

Let p_1 = pressure intensity at section 1-1,

V_1 = velocity of flow at section 1-1,

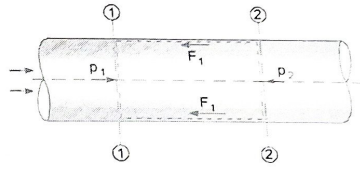
L = length of the pipe between sections 1-1 and 2-2,

d = diameter of pipe,

f' = frictional resistance per unit wetted area per unit velocity,

h_f = loss of head due to friction,

and p_2, V_2 = are values of pressure intensity and velocity at section 2-2.



Uniform horizontal pipe.

Applying Bernoulli's equations between sections 1-1 and 2-2,

Total head at 1-1 = Total head at 2-2 + loss of head due to friction between 1-1 and 2-2

or
$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2 + h_f$$

$z_1 = z_2$ as pipe is horizontal

$V_1 = V_2$ as dia. of pipe is same at 1-1 and 2-2

$$\therefore \quad \frac{p_1}{\rho g} = \frac{p_2}{\rho g} + h_f \text{ or } h_f = \frac{p_1}{\rho g} - \frac{p_2}{\rho g} \qquad ...(i)$$

But h_f is the head lost due to friction and hence intensity of pressure will be reduced in the direction of flow by frictional resistance.

Now frictional resistance = frictional resistance per unit wetted area per unit velocity × wetted area × velocity^2

or
$$F_1 = f' \times \pi dL \times V^2 \quad [\because \text{wetted area} = \pi d \times L, \text{ velocity} = V = V_1 = V_2]$$
$$= f' \times P \times L \times V^2 \quad [\because \pi d = \text{Perimeter} = P] \ ...(ii)$$

The forces acting on the fluid between sections 1-1 and 2-2 are :

1. pressure force at section 1-1 = $p_1 \times A$

where A = Area of pipe

2. pressure force at section 2-2 = $p_2 \times A$
3. frictional force F_1 as shown in Fig.

Resolving all forces in the horizontal direction, we have

$$p_1A - p_2A - F_1 = 0 \qquad ...(10.1)$$

or
$$(p_1 - p_2)A = F_1 = f' \times P \times L \times V^2 \qquad [\because \text{From } (ii),\ F_1 = f'PLV^2]$$

or
$$p_1 - p_2 = \frac{f' \times P \times L \times V^2}{A}$$

But from equation (i), $p_1 - p_2 = \rho g h_f$

Equating the va

or

The forces acting on

Equating the value of $(p_1 - p_2)$, we get

$$\rho g h_f = \frac{f' \times P \times L \times V^2}{A}$$

or

$$h_f = \frac{f'}{\rho g} \times \frac{P}{A} \times L \times V^2 \qquad \ldots(iii)$$

In equation (iii), $\frac{P}{A} = \frac{\text{Wetted perimeter}}{\text{Area}} = \frac{\pi d}{\frac{\pi}{4} d^2} = \frac{4}{d}$

$$\therefore \quad h_f = \frac{f'}{\rho g} \times \frac{4}{d} \times L \times V^2 = \frac{f'}{\rho g} \times \frac{4LV^2}{d} \qquad \ldots(iv)$$

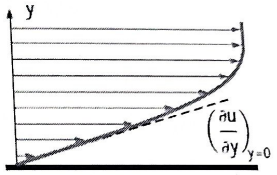
Putting $\frac{f'}{\rho} = \frac{f}{2}$, where f is known as co-efficient of friction.

Equation (iv), becomes as $h_f = \frac{4 \cdot f}{2g} \cdot \frac{LV^2}{d} = \frac{4f \cdot L \cdot V^2}{d \times 2g}$

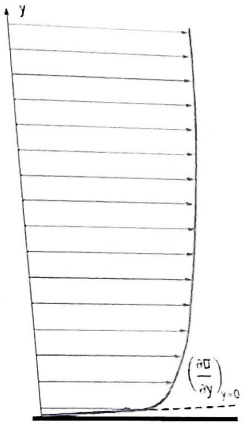
Equation is known as Darcy-Weisbach equation. This equation is commonly used for finding loss of head due to friction in pipes.

Turbulent Flow in Pipes

- The velocity gradients at the wall, and thus the wall shear stress, are much larger for turbulent flow than they are for laminar flow.



Laminar flow

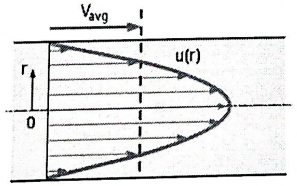


Turbulent flow

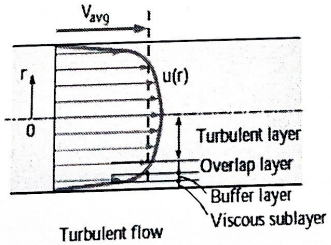
Turbulent Flow in Pipes

Turbulent Velocity Profile

- The velocity profile is parabolic in laminar flow but is much **fuller** in turbulent flow, with a sharp drop near the pipe wall.
- Turbulent flow along a wall can be considered to consist of four regions, characterized by the distance from the wall. The very thin layer next to the wall where viscous effects are dominant is the **viscous** (or **laminar** or **linear** or **wall**) sublayer.



Laminar flow



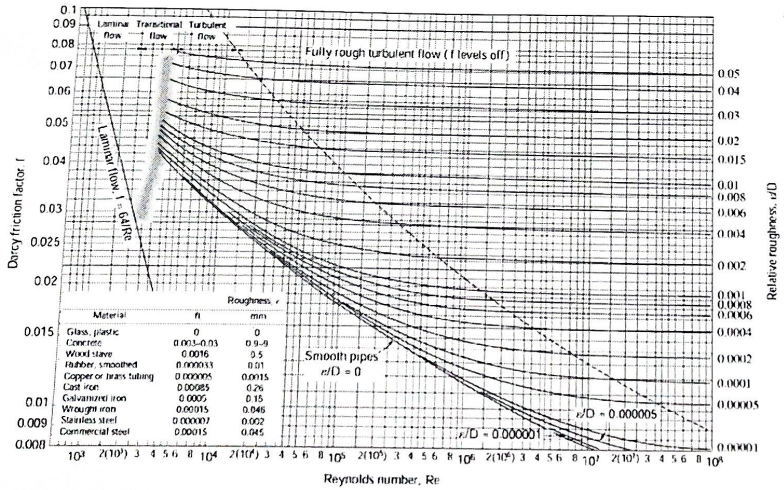
Turbulent flow

The Moody Chart

- The friction factor in fully developed turbulent pipe flow depends on the Reynolds number and the **relative roughness** ε/D, which is the ratio of the mean height of roughness of the pipe to the pipe diameter.
- **Colebrook equation**

$$\frac{1}{\sqrt{f}} = -2.0 \log\left(\frac{\varepsilon/D}{3.7} + \frac{2.51}{Re\sqrt{f}}\right) \qquad \text{(turbulent flow)}$$

- The **Moody chart** presents the Darcy friction factor for pipe flow as a function of the Reynolds number and ε/D over a wide range.
- It is probably one of the most widely accepted and used charts in engineering. Although it is developed for circular pipes, it can also be used for noncircular pipes by replacing the diameter by the hydraulic diameter.



The Moody Chart

- The Moody chart for friction factor for fully developed flow in circular pipes for use in the head loss relation

$$h_L = f\frac{L}{D}\frac{V^2}{2g}.$$

- Friction factors in the turbulent flow are evaluated from the Colebrook equation

$$\frac{1}{\sqrt{f}} = -2\log_{10}\left(\frac{\varepsilon/D}{3.7} + \frac{2.51}{Re\sqrt{f}}\right)$$

The Moody Chart

*Equivalent roughness values for new commercial pipes**

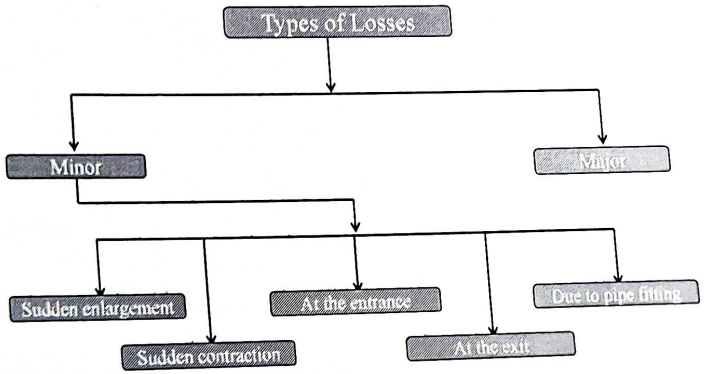
Material	Roughness, ε ft	mm
Glass, plastic	0 (smooth)	
Concrete	0.003–0.03	0.9–9
Wood stave	0.0016	0.5
Rubber, smoothed	0.000033	0.01
Copper or brass tubing	0.000005	0.0015
Cast iron	0.00085	0.26
Galvanized iron	0.0005	0.15
Wrought iron	0.00015	0.046
Stainless steel	0.000007	0.002
Commercial steel	0.00015	0.045

* The uncertainty in these values can be as much as ±60 percent.

Relative Roughness, ε/D	Friction Factor, f
0.0*	0.0119
0.00001	0.0119
0.0001	0.0134
0.0005	0.0172
0.001	0.0199
0.005	0.0305
0.01	0.0380
0.05	0.0716

* Smooth surface. All values are for Re = 10^6 and are calculated from the Colebrook equation.

Types of Losses



Minor

Galvanized iron	0.0005	0.15
Wrought iron	0.00085	0.045
Stainless steel	0.00002	0.045
Commercial steel	0.00015	0.045

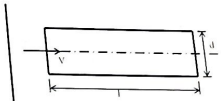
• The uncertainty in these values can be as much as ±60 percent

Minor Losses

- The fluid in a typical piping system passes through various fittings, valves, bends, elbows, tees, inlets, exits, enlargements, and contractions in addition to the pipes.
- These components interrupt the smooth flow of the fluid and cause additional losses because of the flow separation and mixing they induce.
- In a typical system with long pipes, these losses are minor compared to the total head loss in the pipes (the **major losses)** and are called **minor losses.**
- Although this is generally true, in some cases the minor losses may be greater than the major losses. This is the case, for example, in systems with several turns and valves in a short distance.

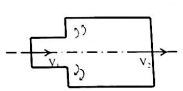
Major loss

→ Loss of head due to friction is termed as major loss.
→ It depends on the velocity of flow, dia. of pipe & roughness.
→ It is denoted as h_f.



$h_f = fLv^2/(2gd)$

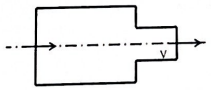
Minor loss



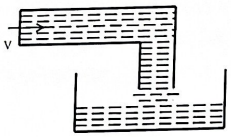
$h_e = (v_1-v_2)^2/(2g)$

→ It is due to distribution in the flow pattern.
i)Loss due to sudden enlargement:-
→ It is due to sudden enlargement of pipe.
→ It is denoted by h_e.

ii)Loss due to sudden contraction:-
→ It is due to sudden contraction of pipe.
→ It is denoted by h_c.



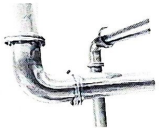
$h_c = [(1/Cc)-1]^2 \times v^2/(2g)$



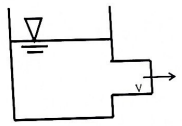
$h_{ex} = v^2/(2g)$

iii)Loss at the exit:-
→ It is head loss at the exit of the pipe.
→ It is denoted by h_{ex}.

iv)Loss due to pipe fittings:-
→ It is due to pipe fittings like elbows, valves etc.
→ It is denoted by h_b.



$h_b = K[v^2/(2g)]$



$h_{en} = 0.5[v^2/(2g)]$

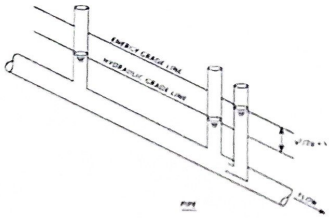
v)Loss at entrance :-
→ It is head loss at the entrance of the pipe.
→ It is denoted by h_{en}.

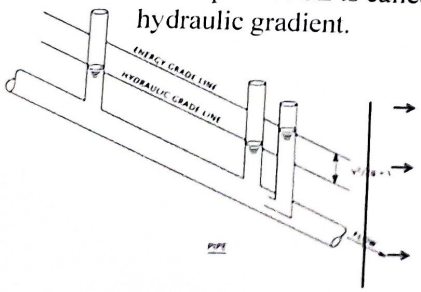
Hydraulic Grade Line

→It is the line joining all the liquid levels in piezometers.

→It is the sum of pressure head & datum head [(P/γ)+z].

→Slope of HGL is called as hydraulic gradient.





Total Energy Line

It is graphical representation of total head at various points along pipe.

It is the sum of pressure head, velocity head & datum head.

[(P/γ)+(v²/2g)+z].

If datum is 0 then energy grade line is TEL.

Pipes in Series

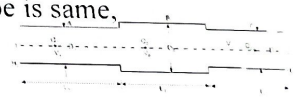
If a pipeline is joined to one or more pipelines in continuation, these are said to constitute pipes in series.

The discharge through the each pipe is same,

$Q_1 = Q_2 = Q_3 = \ldots$

or

$A_1V_1 = A_2V_2 = A_3V_3$



If minor losses are negligible, then the total loss of head is,

$H = h_{f1} + h_{f2} + h_{f3}$

$H = (f_1 l_1 V_1^2/2gd_1) + (f_2 l_2 V_2^2/2gd_2) + (f_3 l_3 V_3^2/2gd_3)$

$H = (f_1 l_1 Q^2/12{*}1d_1^5) + (f_2 l_2 Q_2^2/12{*}1d_2^5) + (f_3 l_3 Q_3^2/12{*}1d_3^5)$

Pipes in Parallel

When a man pipeline divided into two or more parallel pipes it is known as pipes in parallel.

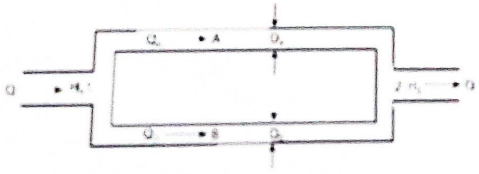
The discharge through the each pipe is,

$Q = Q_1 + Q_2 + Q_3 + \ldots$

and

$h_f = h_{f1} = h_{f2} = \ldots$

We know that $Q = (12{*}1d^5 h_f/fL)^{1/2}$



Substituting this value in Q,

$Q = Q_1 + Q_2 + Q_3 + \ldots$

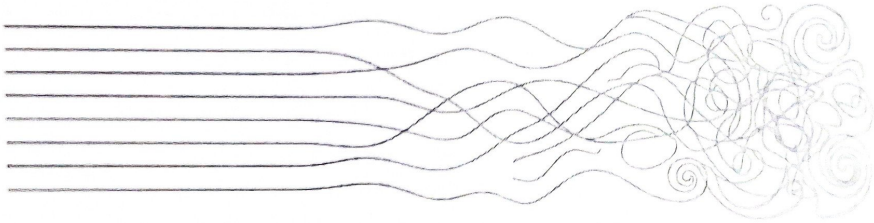
$(12{*}1d_e^5 h_{fe}/f_e L_e)^{1/2} = (12{*}1d_1^5 h_{f1}/f_1 L_1)^{1/2} + (12{*}1d_2^5 h_{f2}/f_2 L_2)^{1/2}$

$+ (12{*}1d_3^5 h_{f3}/f_3 L_3)^{1/2}$

$(d_e^5/f_e L_e)^{1/2} = (d_1^5/f_1 L_1)^{1/2} + (d_2^5/f_2 L_2)^{1/2} + (d_3^5/f_3 L_3)^{1/2}$

Defining the Threshold: The Reynolds Number (Re)

$$\mathrm{Re} = \frac{\rho V D}{\mu}$$



Re < 2300	Re ≈ 2300	High Re	
LAMINAR FLOW Viscous forces dominate. Fluid moves in smooth, parallel layers.	**TRANSITION BENCHMARK** The standard engineering threshold where flow begins to turn chaotic.	**TURBULENT FLOW** Inertial forces dominate. Enhanced mixing, chaotic eddies, and rapid momentum transfer.	The Ideal vs. The Real

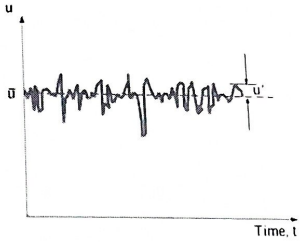
Turbulent Fl

Turbulent Flow in Pipes

- In laminar flow, fluid particles flow in an orderly manner along pathlines, and momentum and energy are transferred across streamlines by molecular diffusion.
- In turbulent flow, the swirling eddies transport mass, momentum, and energy to other regions of flow much more rapidly than molecular diffusion, greatly enhancing mass, momentum, and heat transfer.
- As a result, turbulent flow is associated with much higher values of friction, heat transfer, and mass transfer coefficients.
- The eddy motion in turbulent flow causes significant fluctuations in the values of velocity, temperature, pressure, and even density (in compressible flow).

Turbulent Flow in Pipes

- Fluctuations of the velocity component u with time at a specified location in turbulent flow shown in fig below.



- Instantaneous values of the velocity fluctuate about an average value, which suggests that the velocity can be expressed as the sum of an average value $\bar{u}$ and fluctuating component u',

$$u = \bar{u} + u'$$

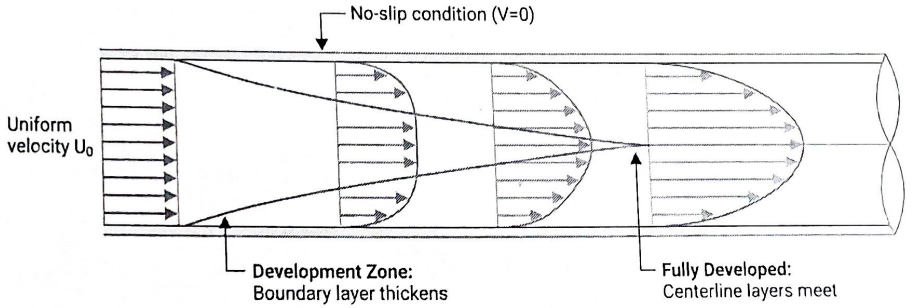
Turbulent Flow in Pipes

Turbulent Shear Stress

- The turbulent shear stress consists of two parts: the laminar component, which accounts for the friction between layers in the flow direction (expressed as $\tau_{lam} = -\mu \, d\bar{u}/dr$) , and the turbulent component, which accounts for the friction between the fluctuating fluid particles and the fluid body (denoted as τ_{turb} and is related to the fluctuation components of velocity).
- Then the total shear stress in turbulent flow can be expressed as

$$\tau_{total} = \tau_{lam} + \tau_{turb}$$

Flow Evolution: The Entrance Region



Laminar Entrance Length
$L / D \approx 0.06\,Re$

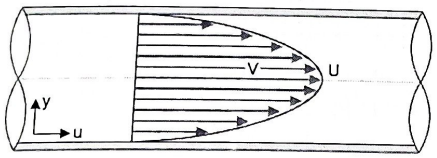
At the transition limit (Re = 2300), this can take up to ~140 pipe diameters to fully develop.

Turbulent Entrance Length
Enhanced mixing causes rapid boundary layer growth.

Mean velocity profile develops within 25 to 40 diameters. Internal turbulent motion may take 80+ diameters to stabilize.

Velocity Profiles: A Structural Comparison

LAMINAR FLOW



Mathematical Ratio

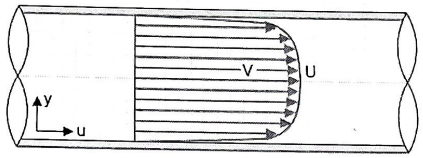
$$\frac{V}{U} = 0.5$$

Average velocity is exactly half the centerline maximum.

Momentum Transfer

Slow transfer via molecular viscosity.

TURBULENT FLOW



Mathematical Ratio

$$\frac{V}{U} \approx 0.8 \text{ to } 0.87$$

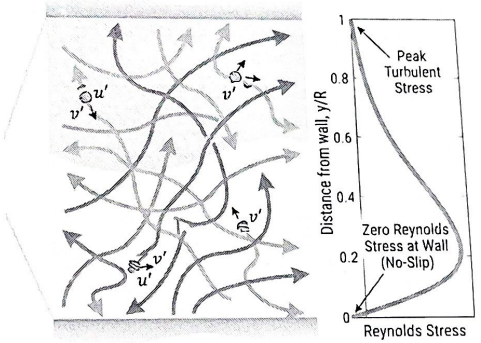
Average velocity is much closer to the maximum.

Momentum Transfer

Rapid momentum transfer between adjacent layers via fluctuating velocity components (u′, v′), drastically flattening the velocity gradient.

The Engine of Turbulence: Reynolds Stress

Apparent stress caused by chaotic particle fluctuations is the true engine of turbulent momentum transfer.



$$\tau = \tau_{lam} + \tau_{turb} = \mu \frac{du}{dy} - \rho(\overline{u'v'})$$

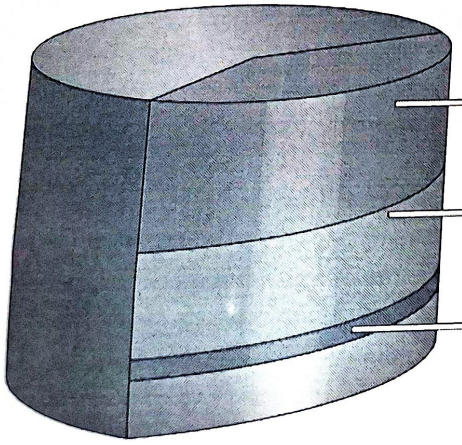
The Fluctuations

u' and v' are random fluctuating velocity components. Because they are negatively correlated, the term $-\rho(\overline{u'v'})$ results in a positive stress addition.

The Wall Paradox

At the exact pipe wall surface, the no-slip condition forces both u' and v' to zero. Turbulent stress vanishes entirely, and **pure viscous shear** completely dominates.

Near-Wall Anatomy: The Structural Layers of Turbulence



Logarithmic Region ($y^+ > 30$)

Turbulent shear fully dominates. Profile follows the empirical curve: $u^+ = 2.5 \ln(y^+) + 5.0$

Buffer Layer ($5 < y^+ < 30$)

The transition zone. Viscous and turbulent shear are actively competing for dominance.

Viscous Sublayer ($0 \le y^+ \le 5$)

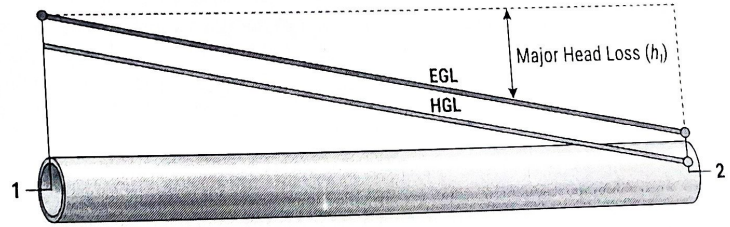
Ultra-thin layer glued to the wall. Viscous shear completely dominates. Velocity profile is strictly linear: $u^+ = y^+$

Dimensionless Wall Coordinate: y^+

Friction Velocity: $u^* = \sqrt{(\tau_w / \rho)}$

System Impact: Major Head Losses

Micro-level wall shear stress translates into macro-level mechanical energy lost permanently as heat.



$$h_l = f \cdot \frac{L}{D} \cdot \frac{V^2}{2g}$$

f : Dimensionless friction factor

$\frac{L}{D}$: Geometric ratio of length to diameter

$\frac{V^2}{2g}$: Dynamic velocity head

Note on Kinetic Energy: For a 1/7th turbulent profile, the kinetic energy coefficient (α) is ≈ 1.06. Being close to unity, engineers approximate $\alpha = 1$ for most turbulent pipe calculations.

Engin

Engineering Models: The Friction Factor Decision Matrix

What is the Flow Regime?

Laminar Flow (Re < 2300)

Exact Analytical Solution

$$f = \frac{64}{\mathrm{Re}}$$

Turbulent Flow (Re > 2300)

Smooth Pipe

Blasius Correlation

$\mathrm{Re} \leq 10^5$

$$f = \frac{0.316}{Re^{0.25}}$$

Simple empirical fit.

General Implicit

Colebrook Equation

$$\frac{1}{\sqrt{f}} = -2.0 \log\left(\frac{e}{3.7} + \frac{2.51}{\mathrm{Re}\sqrt{f}}\right)$$

The industry standard, but requires iteration to solve.

General Explicit

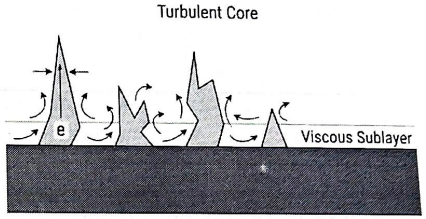
Haaland Approximation

$$\frac{1}{\sqrt{f}} \approx -1.8 \log\left[\left(\frac{e}{3.7}\right)^{1.11} + \frac{6.9}{\mathrm{Re}}\right]$$

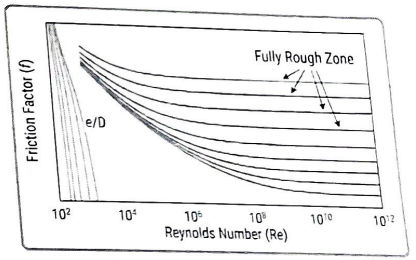
Solves directly; accurate within 2% of Colebrook.

The Asymptotic Limit: The Fully Rough Zone

When relative roughness completely overtakes the fluid dynamics.

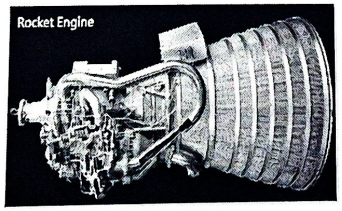


When roughness elements (e) poke through the sublayer, form drag against these elements completely dominates viscous friction.

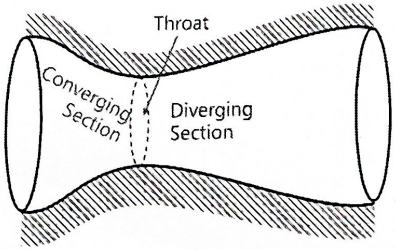


In this 'fully rough' regime, the friction factor (f) ceases to be a function of the Reynolds number. It becomes a constant dictated entirely by the relative roughness (e/D).

Convergent-Divergent Nozzle



- A CD Nozzle is a variable area passage which is used to accelerate gases to higher supersonic speeds.
- It consists of a converging section with minimum area occurring at a specific location called the throat.
- Downstream of the throat, the cross-sectional area starts to increase, thus creating the diverging section of the CD nozzle.
- It is most commonly used in propulsion systems such as rocket engines and after-burners of jet engines.
- It is also used in supersonic wind tunnels.

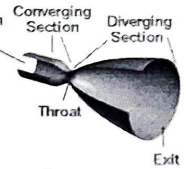


Nozzle

- A nozzle is a relatively simple device, just a specially shaped tube through which hot gases flow.
- Ramjets and rockets typically use a fixed convergent section followed by a fixed divergent section for the design of the nozzle. This nozzle configuration is called a **convergent-divergent**, or **CD**, nozzle.
- In a CD nozzle, the hot exhaust leaves the combustion chamber and converges down to the minimum area, or **throat,** of the nozzle.
- The throat size is chosen to **choke** the flow and set the mass flow rate through the system.
- The flow in the throat is sonic which means the Mach Number is equal to one in the throat.
- Downstream of the throat, the geometry diverges and the flow is isentropically expanded to a supersonic Mach number that depends on the area ratio of the exit to the throat.
- The expansion of a supersonic flow causes the static pressure and temperature to decrease from the throat to the exit, so the amount of the expansion also determines the exit pressure and temperature.
- The exit temperature determines the exit speed of sound, which determines the exit velocity.
- The exit velocity, pressure, and mass flow through the nozzle determines the amount of thrust produced by the nozzle.

$\dot{m}$ = mass flow rate
V = velocity
ρ = density
γ = specific heat ratio

A = area
p = pressure
M = Mach
a = speed of sound



Conservation of Mass: $\dot{m} = \rho V A = \text{constant}$

$$\frac{d\rho}{\rho} + \frac{dV}{V} + \frac{dA}{A} = 0$$

Conservation of Momentum: $\rho V\, dV = -dp$

Isentropic Flow: $\frac{dp}{p} = \gamma \frac{d\rho}{\rho}$ $\qquad dp = a^2\, d\rho$

Combine with Momentum: $-M^2 \frac{dV}{V} = \frac{d\rho}{\rho}$

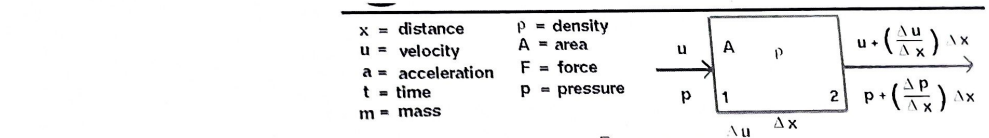
Combine with Mass: $(1 - M^2)\frac{dV}{V} = -\frac{dA}{A}$

For subsonic flow (M < 1), increase in area (dA > 0) causes flow velocity to decrease (dV < 0)

For supersonic flow (M > 1), increase in area (dA > 0) causes flow velocity to increase (dV > 0)

x = distance
u = velocity
a = acceleration
t = time
m = mass

ρ = density
A = area
F = force
p = pressure



$$F = ma = m\frac{\Delta u}{\Delta t}$$

$$-\left[(pA)_2 - (pA)_1\right] = m\frac{u_2 - u_1}{\Delta t}$$

$$-\left[\left(p + \left(\frac{\Delta p}{\Delta x}\right)\Delta x\right)A - pA\right] = m\frac{u + \left(\frac{\Delta u}{\Delta x}\right)\Delta x - u}{\Delta t}$$

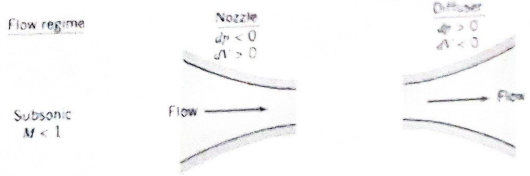
$$m = \rho \Delta x A \qquad -\left(\frac{\Delta p}{\Delta x}\right)\Delta x A = m\left(\frac{\Delta u}{\Delta x}\right)\frac{\Delta x}{\Delta t} \qquad \frac{\Delta x}{\Delta t} = u$$

$$-\frac{\Delta p}{\Delta x} = \rho u \frac{\Delta u}{\Delta x}$$

Differential Form: $-\frac{dp}{dx} = \rho u \frac{du}{dx}$

Subsonic Flow, M < 1

- Case I
- For M < 1, the factor $1/[1 - M^2]$ is positive, so that a positive dA leads to a positive dp and a negative dV.
- These mathematical results mean that in a divergent section (dA > 0)
- Flow must experience an increase in pressure (dp > 0) and the velocity must decrease (dV < 0). Hence a divergent channel is a subsonic diffuser (a diffuser is a device that decelerates a flow).
- Case II
- a negative dA leads to a negative dp and a positive dV.
- These mathematical results mean that in a convergent section (dA < 0) the flow must experience a decrease in pressure (dp < 0) and the velocity must increase (dV > 0).
- Hence a convergent channel is a subsonic nozzle (a nozzle is a device that accelerates a flow).



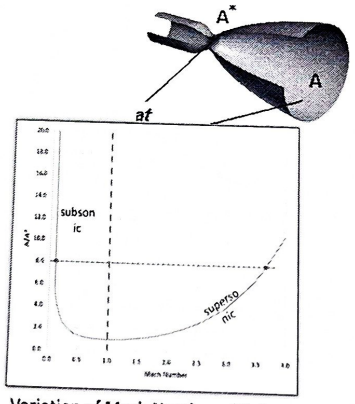
Super-Sonic Flow M > 1

- Case I
- For M > 1, the factor $1/[1 - M^2]$ is negative, so that a positive dA leads to a negative dp and a positive dV. These mathematical results mean that in a divergent section (dA > 0) the flow must experience a decrease in pressure (dp < 0) and the velocity must increase (dV > 0). Hence a divergent channel is a supersonic nozzle.
- Case II
- A negative dA leads to a positive dp and a negative dV.
- These mathematical results mean that in a convergent section (dA < 0) the flow must experience an increase in pressure (dp > 0) and the velocity must decrease (dV < 0).
- Hence a convergent channel is a supersonic diffuser.

- Using the continuity and the isentropic relations, we can get the following equation relating the area to the Mach number:

$$\frac{A}{A^*} = \frac{1}{M}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}}$$

- This equation is called the area – Mach number relation, and it shows that $M = f(A/A^*)$, i.e., the Mach number at any location in the duct is a function of the ratio of the local duct area to the sonic throat area.
- Note that there are two values of M that correspond to a given area ratio (A/A^*), a subsonic and a supersonic value.
- The solution to this equation is plotted in the graph on the right, clearly showing the subsonic and the supersonic branches.



Variation of Mach Number with Area

Critical velocity

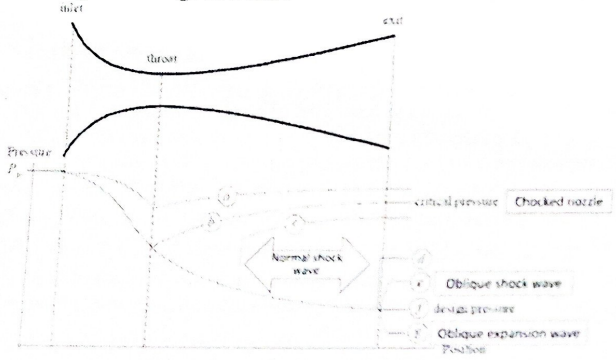
- The critical velocity is the velocity at the throat of a correctly designed convergent-divergent nozzle, or the velocity at the exit of a convergent nozzle when the pressure ratio across the nozzle is the critical pressure ratio or less.

$$V_c = \sqrt{\gamma \cdot R \cdot T_c}$$

$$\text{Critical Temperature Ratio} = \frac{T_c}{T_1} = \frac{2}{\gamma+1}$$

$$\text{Critical PressureRatio} = \pi_c = \frac{P_c}{P_1} = \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}}$$

Convergent-divergent nozzle



Convergent-divergent nozzle

- When the discharge pressure is slightly lower than the inlet pressure (case-a), the flow is subsonic throughout the nozzle. In this case, the lowest pressure and highest velocity occur at the nozzle throat where the cross-sectional area reaches a minimum.
- As the discharge pressure is decreased (critical back pressure), the velocity at the throat continues to increase until eventually it reaches the local speed of sound at the throat (case-b). At this conditions the nozzle is chocked and the mass flow rate reaches its maximum value.
- As the discharge pressure is reduced still further (case-c), the mass flow rate through the nozzle remains constant but the flow accelerates from the speed of sound at the throat to supersonic velocities in the diverging section of the nozzle At some point downstream of the throat, the flow suddenly changes from being supersonic to being subsonic (shock wave).

Convergent-divergent nozzle

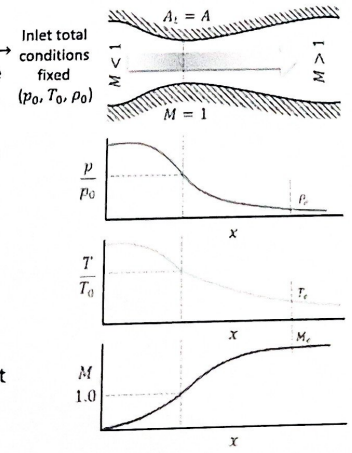
- The shock wave occurring inside the nozzle is called a normal shock because it is perpendicular to the flow direction. **Note** that shock waves can only occur at locations where the upstream flow is supersonic and the downstream of the shock wave, the flow must be subsonic.
- The location of the normal shock moves towards the nozzle exit as the discharge pressure is further reduced. (Case-d) illustrates the behavior for the back pressure value that causes a shock wave exactly at the nozzle exit.
- If the back pressure is decreased still further, an oblique shock wave occurs outside of the nozzle as the pressure adjusts from the supersonic condition at the nozzle exit to the imposed back pressure (case-e). The oblique shock wave will, in general, not be perpendicular to the flow. This condition of a discharge pressure that is lower than the imposed back pressure is referred to as overexpansion.

Convergent-divergent nozzle

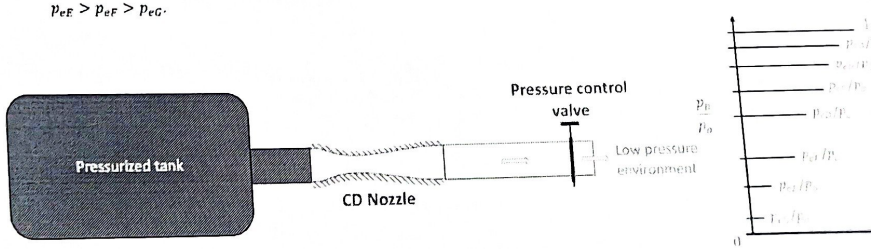
- When the back pressure is reduced to the design pressure (case-f), supersonic flow leaves the nozzle with an exit pressure that is exactly equal to the applied back pressure and no shock occurs. If the back pressure is reduced to a value that is below the design pressure (case-g), an oblique expansion wave results. This behavior is not a pressure discontinuity but rather a continuous wave that is a result of the flow exiting the nozzle coming to equilibrium with the low external pressure. This situation is called under-expansion.

1D Flow through a CD Nozzle

- Consider a CD nozzle such that the area ratio at the inlet $A_i/A^* \rightarrow \infty$. The inlet station feeds from a large reservoir operating at the p_0, ρ_0 and T_0 (stagnation properties).
- In the convergent portion of the nozzle, the subsonic flow is accelerated, and the Mach number is dictated by the local value of A/A^*. At the throat, where $A_t = A^*$, we get $M = 1$.
- In the divergent portion of the nozzle the flow expands supersonically, and again the Mach number (supersonic now) is governed by the local value of A/A^*.
- The resulting variations of the pressure, temperature and Mach number follow a monotonic increase or decrease as shown on the right.
- P_e, T_e and M_e are the pressure, temperature and Mach number at the nozzle exit. In an ideal situation the exit pressure P_e is equal to the ambient pressure at the exit (back pressure, p_b) – this situation is referred to as "design condition."



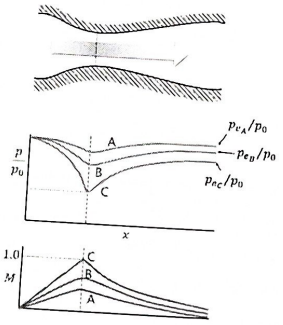
- For our analysis, we will use the physical model of the CD nozzle as shown in the figure.
- The nozzle is connected to a tank supplying the inlet station with fixed total pressure and temperature.
- A valve attached to the nozzle exit is adjusted to vary back pressure.
- Except for two special conditions which we will discuss later, the nozzle exit pressure will always be equal to the back pressure. For the purpose of our discussion, we will be using the two terms interchangeably and make the distinction when necessary.
- Let us analyze what happens to the flow as we open the valve and reduce the back pressure from $p_{eA} > p_{eB} > p_{eC} > p_{eD} > p_{eE} > p_{eF} > p_{eG}$.



- P_e, T_e and M_e are
the nozzle exit. In an ide
to the ambient pressure at the
situation is referred to as "design co

Fully Subsonic Flow

- Initially, $p_e = p_b = p_0$. Therefore, there will be no flow in the nozzle.
- When the back pressure is slightly reduced such that $p_e = p_b = p_{eA}$, a low-speed flow is established in the nozzle, as indicated by Case A.
- As we start decreasing the back/exit pressure, p_b/p_e, from p_{eA} to p_{eB}, this flow starts accelerating as shown by case B.
- If the flow in the entire nozzle is subsonic, then the flow at any axial station within the nozzle can be analyzed using the relations derived previously.
- As the exit pressure is further decreased, the flow is accelerated more, leading to a higher Mach number throughout the nozzle with the highest being at the throat. Consequently, the pressure is also lowest at the throat.
- At a certain value of $p_b = p_e = p_{eC}$, the flow just becomes sonic at the throat (Case C). At this point $A_t = A^*$ and $M_t = 1$.

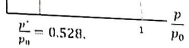


Choked Flow – Converging Section

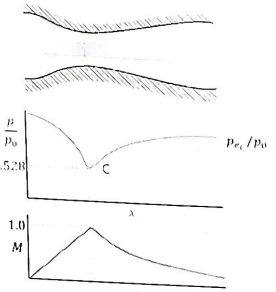
- Decreasing the exit pressure from p_{eA} to p_{eC}, the mass flow through the nozzle ($\dot{m} = \rho_t A_t V_t$) increases until sonic conditions are reached at the throat corresponding to p_{eC}.
- The mass flow rate through the CD nozzle corresponding to condition C, i.e., sonic flow attained at the throat, is given by:

$$\dot{m} = \rho_t A_t V_t = \rho^* A^* V^*$$

- If the back pressure is further reduced, the Mach number at the throat cannot increase beyond 1.0 and the mass flow remains constant. This condition is referred to as choked flow.



- For $\gamma = 1.4$, the sonic flow at the throat corresponds to the pressure ratio of $p^*/p_0 = 0.528$.

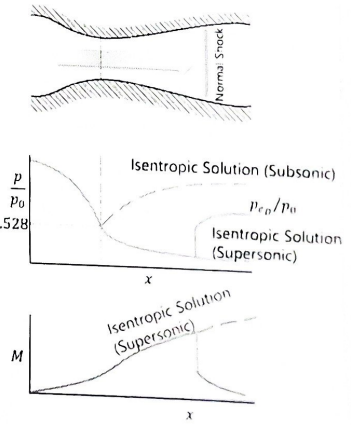


How Choking Works:

- **How Choking Works:**
- **Subsonic Flow (Converging Section)**: As the gas flows through the converging section of the nozzle, its velocity increases, and the pressure decreases.
- **At the Throat (Choking Condition)**: When the flow reaches the throat, if the conditions are right (specifically if the pressure difference between the inlet and the exit is large enough), the flow will reach Mach 1, and the flow becomes choked. This means that no further increase in mass flow rate can occur with an increase in downstream pressure; the mass flow rate is now limited by the conditions at the throat.
- **Supersonic Flow (Diverging Section)**: After the throat, if the nozzle continues into a diverging section, the flow can accelerate to supersonic speeds (Mach > 1) if the pressure ratio is sufficiently high.

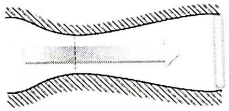
Choked Flow – Diverging Section

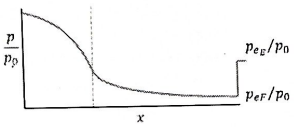
- If the exit pressure is further decreased to a value below the sonic condition, $p_{eD} < p_{eC}$, the flow remains unchanged in the converging section. However, an interesting flow phenomenon occurs in the diverging section of the nozzle.
- No isentropic solution is possible in the divergent duct until the nozzle exit pressure is adjusted to the specified low value as shown earlier for the design condition case.
- For values of exit pressure above the supersonic design condition value, but below p_{e_C}(sonic condition), a normal shock wave is observed inside the diverging section as shown.
- The region ahead of the shock in the diverging section is supersonic, while behind the shock the flow is subsonic. As a result, the Mach number decreases and the static pressure increases toward the exit.

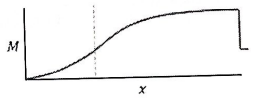


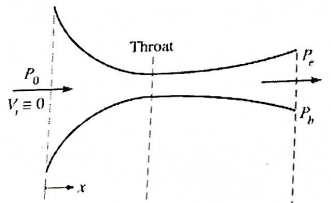
Choked Flow – Diverging Section (cont.)

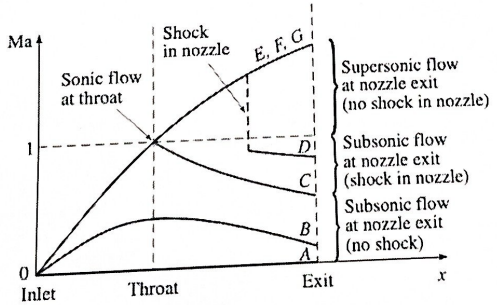
- The location of the normal shock is such that it allows the exit pressure to attain the prescribed value $p_{e_D} = p_b = p_e$ at the nozzle exit.
- The flow accomplishes this via the increase of static pressure across the shock wave and the pressure increase due to the subsonic flow in the remaining diverging section.
- As we decrease the exit pressure further from p_{eD}, the shock wave moves downstream and closer to the nozzle exit.
- At a certain value $p_{e_E} < p_{eD}$ the shock will be located precisely at the exit.
- Note that in this figure p_{eF} represents the proper isentropic value for the design exit Mach number and it exists just upstream of the normal shock.
- From this condition, the back pressure and exit pressure will be distinct.
- Now let's reduce the backpressure even further and see what happens to the shock at the exit.

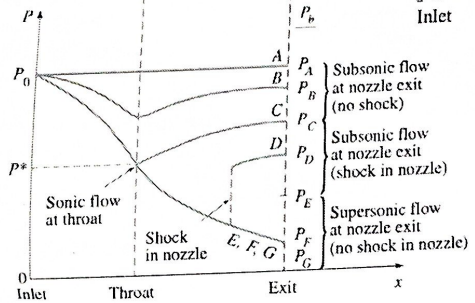












Flow through CD Nozzle

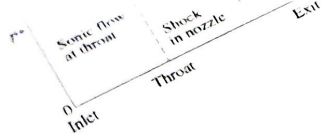
- Consider the converging–diverging nozzle shown in Fig.
- A fluid enters the nozzle with a low velocity at stagnation pressure P_0.
- When $P_b = P_0$ (case *A*), there is no flow through the nozzle. since the flow in a nozzle is driven by the pressure difference between the nozzle inlet and the exit.
- Different Cases when back pressure is lowered.

- Case A
  - When *P*0 . *Pb* . *PC*, the flow remains subsonic throughout the nozzle, and the mass flow is less than that for choked flow.
  - The fluid velocity increases in the first (converging) section and reaches a maximum at the throat (but Ma =< 1).
  - However, most of the gain in velocity is lost in the second (diverging) section of the nozzle, which acts as a diffuser.
  - The pressure decreases in the converging section, reaches a minimum at the throat, and increases at the expense of velocity in the diverging section.

Case B

- When *Pb* = *PC*, the throat pressure becomes *P** and the fluid achieves sonic velocity at the throat.
- But the diverging section of the nozzle still acts as a diffuser, slowing the fluid to subsonic velocities.
- The mass flow rate that was increasing with decreasing *Pb* also reaches its maximum value.
- Recall that *P** is the lowest pressure that can be obtained at the throat, and the sonic velocity is the highest velocity that can be achieved with a converging nozzle.
- Thus, lowering *Pb* further has no influence on the fluid flow in the converging part of the nozzle or the mass flow rate through the nozzle. However, it does influence the character of the flow in the diverging section.

- Case C

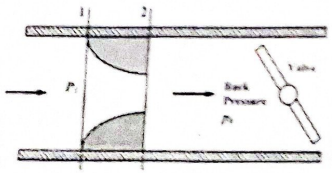


- ***Case C***
- **When *PC* . *Pb* . *PE*, the fluid that achieved a sonic velocity at the throat continues accelerating to supersonic velocities in the diverging section as the pressure decreases. This acceleration comes to a sudden stop, however, as a normal shock develops at a section between the throat and the exit plane, which causes a sudden drop in velocity to subsonic levels and a sudden increase in pressure. The fluid then continues to decelerate further** in the remaining part of the converging–diverging nozzle.
- Flow through the shock is highly irreversible, and thus it cannot be approximated as isentropic. The normal shock moves downstream away from the throat as *Pb* is decreased, and it approaches the nozzle exit plane as *Pb* approaches *PE*.
- When *Pb* = *PE*, the normal shock forms at the exit plane of the nozzle. The flow is supersonic through the entire diverging section in this case, and it can be approximated as isentropic.
- However, the fluid velocity drops to subsonic levels just before leaving the nozzle as it crosses the normal shock.

- Case D
- When *P* > *Pb* > 0, the flow in the diverging section is supersonic, and the fluid expands to *PF* at the nozzle exit with no normal shock forming within the nozzle. Thus, the flow through the nozzle can be approximated as isentropic.
- When *Pb* = *PF*, no shocks occur within or outside the nozzle.
- When *Pb* < *PF*, irreversible mixing and expansion waves occur downstream of the exit plane of the nozzle
- When *Pb* > *PF*, however, the pressure of the fluid increases from *PF* to *Pb* irreversibly in the wake of the nozzle exit, creating what are called *oblique shocks*.

Maximum flow rate through the nozzle

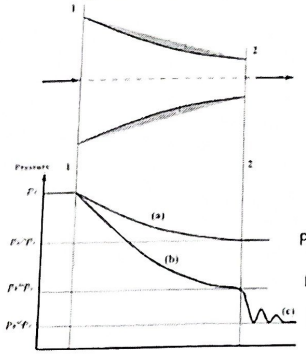
- When the back pressure P_b, is equal to the inlet pressure P_1, then there no fluid flow.
- As P_b reduces the mass flow rate through the nozzle increases.
- When the back pressure reaches the critical value p_c, then the velocity at exit of the convergent nozzle is sonic and the mass flow through the nozzle is maximum and the nozzle is said to be chocked.



Maximum flow rate through the nozzle

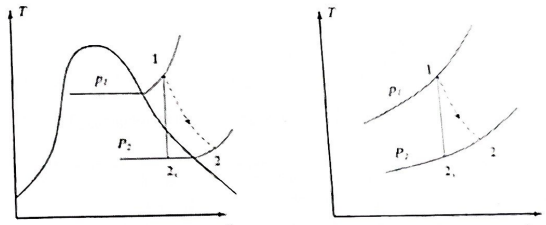
- Further reduction of the back pressure below the critical value dose not affect the mass flow which remains at its maximum value. The exit pressure remains at p_c, and expansion to the back pressure occurs outside the nozzle.
- **For a convergent nozzle**, the maximum mass flow is obtained when the pressure ratio across the nozzle is the critical pressure ratio
- **For convergent-divergent nozzle**, the flow is maximum and fixed when the pressure at the throat is critical pressure.

Convergent nozzle



$P_b > Pc$	Over-expansion	$m < m_{max}$
$P_b = Pc$	Chocked Nozzle	$m = m_{max}$
$P_b < Pc$	Under-expansion	$m = m_{max}$

Nozzle efficiency



$$\eta = \frac{V_2^2 - V_1^2}{V_{2s}^2 - V_1^2} = \frac{h_1 - h_2}{h_1 - h_{2s}}$$

Nozzle efficiency

- *The ideal process for the expansion in the nozzle is the reversible adiabatic (isentropic) process. Due to frication either between the fluid and the walls of the nozzle or within the fluid itself, the expansion process is irreversible but still adiabatic.*
- *In nozzle design it is common to build all calculations on isentropic flow and then taking the frication into consideration through an efficiency called the adiabatic or the isentropic efficiency.*

Calculation coefficients.

- **Nozzle velocity coefficient CV.**
- The nozzle velocity coefficient is defined as the ratio of the actual velocity at exit from the nozzle to the velocity at exit when the flow is isentropic (ideal).

$$CV = \frac{V_2}{V_{2s}}$$

- **Nozzle coefficient of discharge Cd.**
- The coefficient of discharge of a nozzle is defined as the ratio of the actual mass flow through the nozzle (m), to the mass flow through the nozzle if the flow is isentropic (m_s).

$$Cd = \frac{m}{m_s}$$

- Gases expand in a propulsion nozzle from 3.5 bar and 425°C down to a back pressure of 0.97 bar at the rate of 18 kg/s. Taking a coefficient of discharge of 0.99 and a nozzle efficiency of 0.94, calculate the required throat and exit areas of the nozzle. For the gases take Cp = 1.11 KJ/Kg and ɣ = 1.333. Assume that the inlet velocity is negligible.

Example

- Gases expand in a propulsion nozzle from 3.5 bar and 425°C down to a back pressure of 0.97 bar at the rate of 18 kg/s. Taking a coefficient of discharge of 0.99 and a nozzle efficiency of 0.94, calculate the required throat and exit areas of the nozzle. For the gases take Cp = 1.11 KJ/Kg and ɣ = 1.333. Assume that the inlet velocity is negligible.
- Solution: The critical pressure ratio and the corresponding critical pressure are:

$$\frac{P_c}{P_1} = \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}} = \left(\frac{2}{2.333}\right)^{\frac{1.333}{0.333}} = 0.541$$

$$P_c = 3.5 \times 0.541 = 1.894\,(bar)$$

- Since $P_b < P_c$, then the nozzle is chocked.

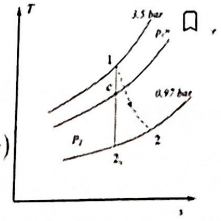
Example

$$\frac{T_c}{T_1} = \frac{2}{\gamma+1} = \frac{2}{2.333} = 0.857$$

$$T_c = (425+273)\times 0.857 = 598\,(K)$$

$$V_c = \sqrt{\gamma RT} = \sqrt{1.11\times 0.2775\times 10^3 \times 598} = 471\,(m/s)$$

$$v_c = \frac{RT_c}{P_c} = \frac{0.277\times 598}{1.984\times 100} = 0.875\,(m^3/kg)$$



- All these calculations based on the isentropic assumption.

Example

- Based on the coefficient of discharge we can find the isentropic mass flow rate as:

$$Cd = \frac{m}{m_s}$$

$$0.999 = \frac{18}{m_s} \qquad m_s = 18.18\,(kg/s)$$

$$m_s = \frac{A_c V_c}{v_c}$$

$$A_c = \frac{18.18\times 0.875}{471} = 0.0338\,(m^2)$$

Example

- Calculate the area at the exit section:

$$\frac{T_{2s}}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$T_{2s} = 698 \times \left(\frac{0.97}{3.5}\right)^{\frac{0.333}{1.333}} = 506(K)$$

- Apply isentropic efficiency:

$$\eta = \frac{T_1 - T_2}{T_1 - T_{2s}} \qquad \therefore 0.94 = \frac{698 - T_2}{698 - 506}$$

$$T_2 = 517.5(K)$$

Example

$$v_2 = \frac{RT_2}{P_2} = \frac{0.2775 \times 517.5}{0.97 \times 100} = 1.48(m^3/kg)$$

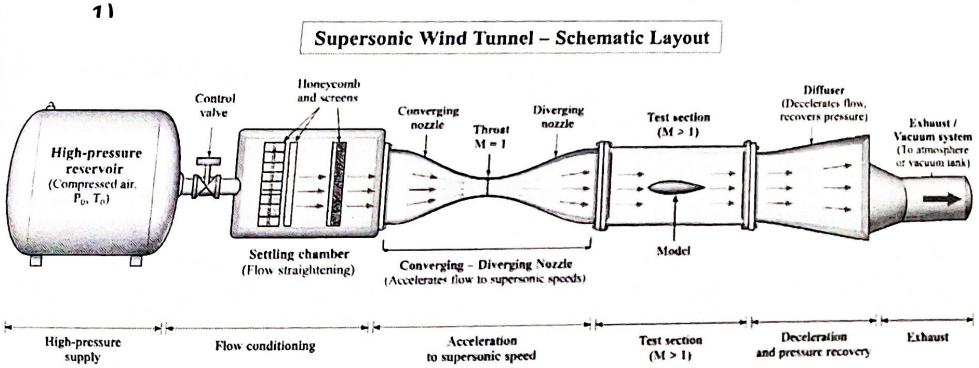
$$V_2 = 44.72\sqrt{C_p(T_1 - T_2)} = 44.72\sqrt{1.11\times10^3 \times (698 - 517.5)} = 633(m/s)$$

$$m = \frac{A_2 V_2}{v_2}$$

$$A_2 = \frac{18\times1.48}{633} = 0.0422(m^2)$$

Supersonic wind tunnel

- *Test facility used to study the aerodynamic behavior of aircraft, missiles, projectiles, and other bodies when the airflow is* **supersonic (Mach > 1)**



High-pressure reservoir

- The air is initially stored in a **high-pressure reservoir**.
- The reservoir contains compressed air at high **stagnation pressure P0** and **stagnation temperature T0**.
- The air has relatively low velocity inside the large reservoir.
- The high pressure provides the energy required to accelerate the air through the tunnel.
- A **control valve** regulates the amount of air entering the tunnel.
- The reservoir pressure is much higher than the pressure in the test section.
- P0>Ptest This pressure difference drives the flow.

Settling chamb

- After passing through
- Its purpose is to
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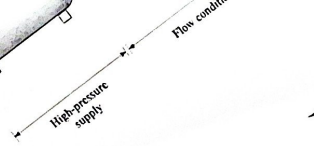

Settling chamber

- After passing through the control valve, air enters the **settling chamber**.
- Its purpose is to produce a **uniform, low-turbulence flow** before the air enters the nozzle.
- The figure shows:
  - **Honeycomb**
  - **Screens**
- **Honeycomb**
- The honeycomb consists of many small passages.
- It:
  - straightens the flow,
  - removes large-scale transverse velocity components,
  - reduces flow angularity.
- **Screens**
  - Screens further reduce turbulence and improve flow uniformity.
  - Thus, the settling chamber prepares the air for the nozzle.
- **Why is this important?**
- If the flow entering the nozzle is highly turbulent or non-uniform, the flow in the test section will also be non-uniform, making experimental measurements inaccurate.

Converging nozzle

- The conditioned air now enters the **converging portion of the nozzle**.
- The cross-sectional area decreases:

$$A\downarrow$$

- For subsonic flow through a converging nozzle:

$$V\uparrow$$

- therefore

$$M\uparrow$$

- The air accelerates as it approaches the throat.

Throat (M=1)

- At the design operating condition, the flow becomes **choked** at the throat:
- M=1
- At the throat:

$$V=a$$

where a is the local speed of sound.

- **What does "choked" mean?**
- Once the throat reaches M=1, the mass flow rate reaches its maximum value for the given upstream stagnation conditions and throat area.
- Further reduction of downstream pressure does **not** make the throat flow faster than M=1.
- Instead, the flow becomes **supersonic downstream of the throat**.

Diverging nozzle

- Immediately after the throat, the nozzle begins to **diverge**:

$$A\uparrow$$

- For **supersonic flow**, increasing area causes the flow to accelerate further:

$$A\uparrow \Rightarrow M\uparrow$$

- Therefore:

M=1 at throat

and downstream:

M>1

Test section — M>1

- After the diverging nozzle, the air enters the **test section**.
- This is the region where the aerodynamic model is placed.
- For example:
  - Aircraft wing
  - Missile
  - Projectile
  - Airfoil
  - UAV configuration
  - Supersonic inlet
  - Nozzle geometry
- The objective is to reproduce the required supersonic flow conditions around the model.
- For example:
- $M_{\infty}=2$, means that the freestream flow around the model is approximately twice the local speed of sound.

Diffuser

- After passing over the model, the high-speed air enters the **diffuser**.
- The diffuser has a very important job:
  - **Reduce the flow velocity and recover pressure before the air leaves the tunnel.**
- The figure shows the diffuser expanding in area.
- However, there is an important difference here.
- **In the supersonic test section:**

  $A\uparrow \Rightarrow V\uparrow$ ⇒ because the flow is supersonic.
- **In a diffuser after the flow has become subsonic:**

  $A\uparrow \Rightarrow V\downarrow$
- So the diffuser is designed to eventually **decelerate the flow and increase its static pressure**.